

Client Segmentation

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1. Project Overview

- **Project Title:** Bank Customer Financial Behavior Clustering.
 - **Description:** This project utilizes unsupervised machine learning to group bank customers based on their financial and demographic attributes.
 - **Problem Statement:** Financial institutions often struggle to provide personalized marketing plans for a diverse customer base, leading to inefficient campaigns.
 - **Objective:** To group customers into distinct clusters that share similar financial and behavioral characteristics to design suitable marketing plans.
 - **Type of ML Task:** Unsupervised Learning (Clustering).
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2. Business / Real-World Context

- **Stakeholders:** Marketing and Sales departments within the bank.
- **Decision Support:** Helps marketing teams decide which product offers (e.g., loans, savings plans) are most relevant to specific customer groups.
- **Risk of Wrong Predictions:** * **Inaccuracy:** Targeting the wrong demographic results in wasted marketing budget and customer annoyance.

3. Dataset Description:

- **Data Source:** bank-full.csv (Standard banking campaign dataset).
 - **Size:** 45,211 rows and 17 features.
 - **Target:** Unsupervised task (no target), though the y variable (subscription) is available for validation.
 - **Key Features:**
 - **Demographic:** Age, Job, Marital Status, Education.
 - **Financial:** Balance, Default, Housing, Loan.
 - **Campaign:** Contact type, Duration, Campaign frequency, Poutcome.
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4. Data Preprocessing

- **Exploration of Unique Values:** Identified 12 job titles, 4 education levels, and 3 marital statuses.
- **Cleaning:** Analyzed for negative ages (none found) and negative balances (3,766 instances identified).
- **Categorical Handling:** Preparation for encoding categorical variables such as job, marital, and education.

5. Exploratory Data Analysis (EDA)

- **Distribution Analysis:** Used boxplots for numerical features (like balance and age) and count plots for categorical features (like job and marital) to understand data spread and imbalances.
 - **Key Statistics:** * The average age is approximately 41 years.
 - Average yearly balance is 1,362 Euros, but with high standard deviation (3,044), indicating significant wealth disparity.
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6. Modeling Approach

- **Clustering Technique:** K-Means Clustering (implied by the cluster outputs in the notebook).
- **Dimensionality Reduction:** Used **PCA (Principal Component Analysis)** to reduce data to two dimensions (PCA1, PCA2) for visualization purposes.
- **Visual Strategy:** Used `sns.scatterplot` with PCA components to visualize the separation between clusters.

7. Results & Insights

Based on the analysis, the project identified 5 distinct customer segments:

- **Cluster 0:** Stable Working-Class Married Customers.
 - **Cluster 1:** Young Single Professionals.
 - **Cluster 2:** High-Value Married Managers.
 - **Cluster 3:** Students & Early-Career Prospects.
 - **Cluster 4:** Affluent Retired Customers.
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8. Technologies & Tools

- **Language:** Python
- **Data Manipulation:** Pandas, NumPy.
- **Visualization:** Matplotlib, Seaborn.
- **Machine Learning:** Scikit-learn (for PCA and Clustering).

9 Model selection:

For this we used 3 different ML techniques and selected one of them.

1. DB-Scan.
2. Hierarchical Silhouette Score.
3. K-Means.

We selected between them based on the Calculate the Silhouette Score and logically for each model

- Calculate the Silhouette Score:

K-Means Silhouette Score: 0.4404

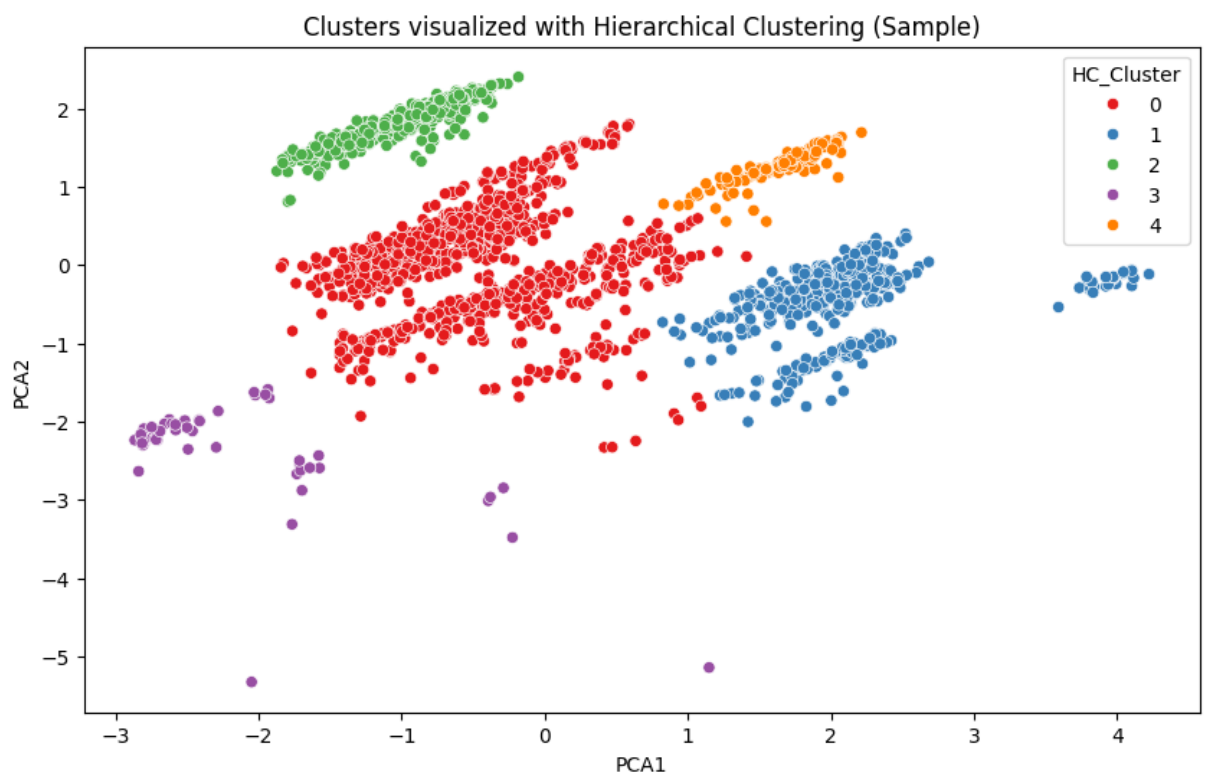
DBSCAN Silhouette Score (excluding noise): 0.0997

Hierarchical Silhouette Score (on sample): 0.5125

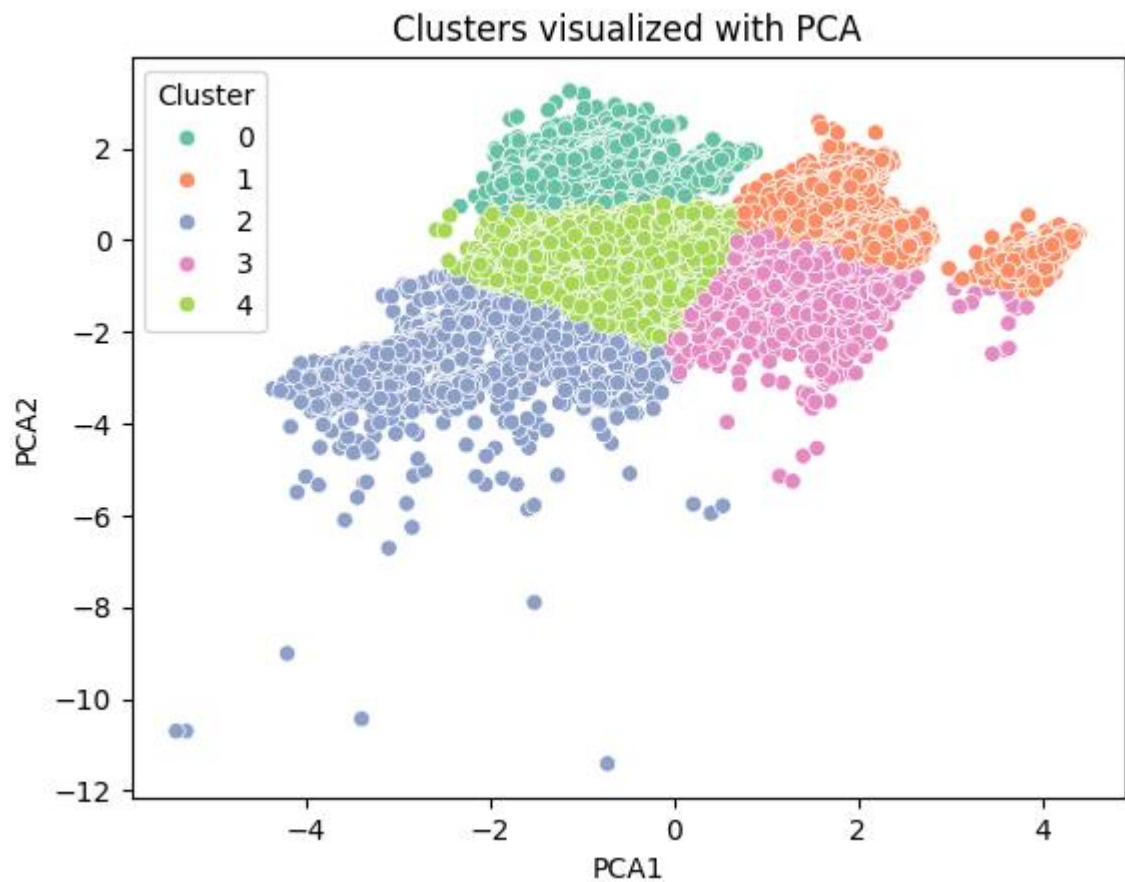
For this it's clear that the Hierarchical Silhouette Score is the highest but wait we can look at the PCA visualization for Hierarchical and K-Means

- PCA's:

Hierarchical



K-Means:



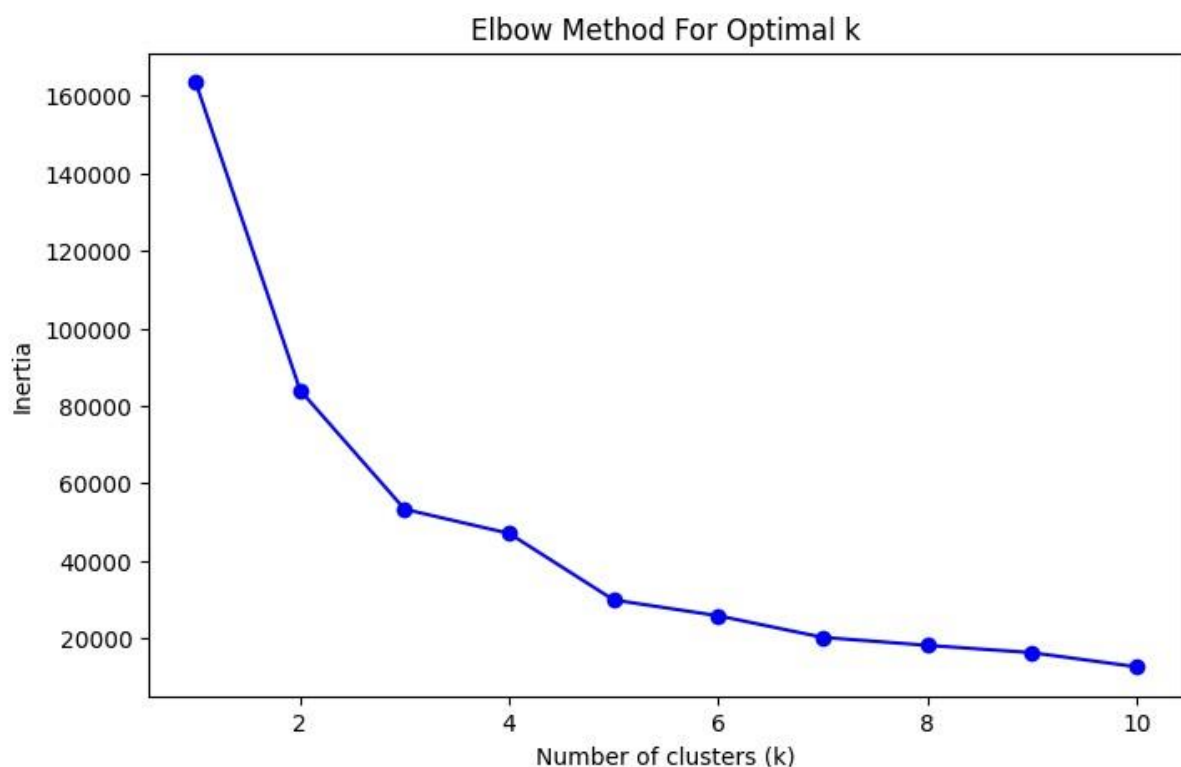
Since the Hierarchical is clustering based on only a sample of the data so it gave higher Hierarchical Silhouette Score but in visualization it seems that the K-Means Is better .

So we select K-Means

10. Modeling Approach (Selection of K)

Before finalizing the model, the **Elbow Method** was used to determine the optimal number of clusters (k). We calculated the **Inertia** (Within-Cluster Sum of Squares) for k values ranging from 1 to 10.

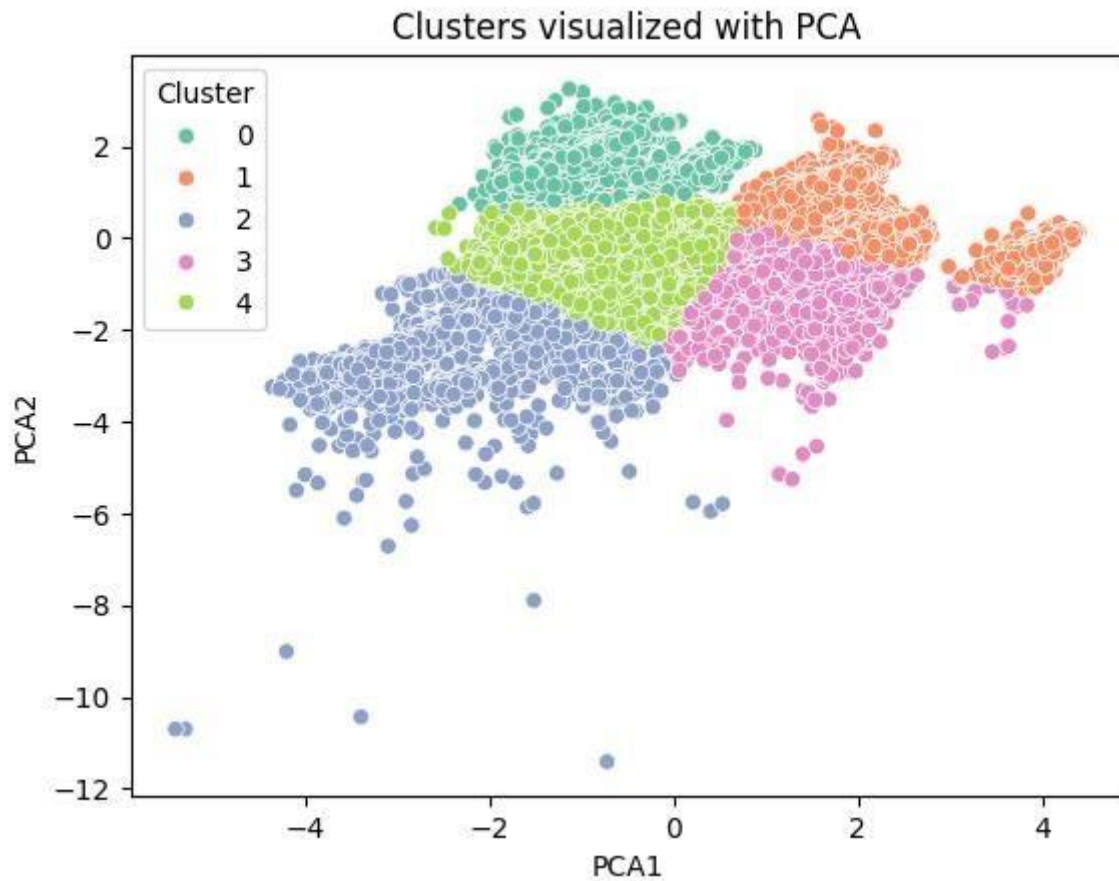
- **Observation:** As seen in the plot below, the "elbow" or the point where the rate of decrease in inertia significantly slows down occurs at $k=5$.
- **Decision:** $k=5$ was selected to provide a granular yet manageable segmentation for the marketing team.



11. Cluster Profiles & Business Interpretation

To justify the segment names and understand the financial behavior of each group, we analyzed the **centroid (mean values)** of each cluster.

Cluster	Avg. Age	Avg. Balance (€)	Job Characteristics	Marital Status	Suggested Segment Label
0	40.8	877.8	Mostly Blue-collar (91%)	Mostly Married (91%)	Stable Working-Class
1	30.5	722.2	High mix of Services/Tech	Single (96%)	Young Single Professionals
2	60.9	4,139.0	Mostly Retired (81%)	Married (80%)	Affluent Retired / High-Value
3	37.0	1,839.3	Management (52%)	Single (87%)	Young Managers / High Earners
4	43.2	1,253.3	Management/Technicians	Married (85%)	Mid-age Professionals



12. How to Run the Project

1. Ensure bank-full.csv is in the project root directory.
2. Open terminal and run:
 - Create virtual environment.
 - Run cd directory to the project root.
 - Activate the environment (virtual environment name\Scripts\activate).
 - Run streamlit run streamlit_file_name.py